Devansh Garg

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Research Interests

Systems for large-scale machine learning: improving the efficiency of training and inference. Current work in datacenter congestion control, RDMA, and programmable data planes. Other areas: in-network computing.

Education

IIT Hyderabad, M.Tech. Computer Science & Engineering — Jul 2025 – May 2028

- **Coursework:** Advanced Data Structures, Software Verification, Security in Intelligent Systems, Visual Computing, Topics in Networks, Systems for ML, Practical Malware Analysis — CGPA: 9.2/10.0

NIT Goa, B.Tech. Computer Science & Engineering — Nov 2021 – May 2025

- **Coursework:** Computer Networks, Computer Organization and Architecture, Theory of Computation — CGPA: 8.98/10.0

IIT Hyderabad, Final-Year CSE Exchange Student — Aug 2024 – May 2025

- **Coursework:** Advanced Topics in Cryptology, Network Security, Quantum Computing, Machine Learning

Publications

Detecting Manipulation to Table Rules in the Programmable Data Planes
Ranjitha K, Karuturi Havya Sree, **Devansh Garg**, Stavan Nilesh Christian, Dheekshitha Bheemanath, Rinku Shah, Praveen Tammana
IFIP Networking 2025, Limassol, Cyprus. pp. 438–446. [PDF]

XF-SISS: XOR-Based Fault Tolerant and Flexible Secret Image Sharing Scheme
Devansh Garg, B. R. Purushothama
IEEE OCIT 2025. pp. 81–86. [IEEE Xplore]

Manuscripts Under Review

Detecting Manipulation to Table Rules in the Programmable Data Planes
Ranjitha K, **Devansh Garg**, Rinku Shah, Praveen Tammana
IEEE Transactions on Network and Service Management (TNSM) — submitted August 2026, under review.
Extended version of the IFIP Networking 2025 paper above.

Research Experience

Graduate Researcher, IIT Hyderabad — advised by Dr. Praveen Tammana — Jul 2025 – Present

- Software-systems-level optimization for AI inference, targeting the cost break-even point at which self-hosting open-weight models becomes cheaper than commercial API access.
- Datacenter congestion control for high-performance networking, using DPDK and RDMA.

Dedicated Dense-Sum Unit for In-Network Allreduce, IIT Hyderabad — course project — Oct 2026 – Present

- Designing a fixed-function path for dense FP16/FP32 sums to extend Flare (SC '21), a programmable RISC-V switch for in-network allreduce, keeping its cores for sparse data and custom operators.
- Reproduced Flare in the PsPIN cycle-accurate simulator, matching 52 of 56 configurations within $\pm 4\%$ of published results.
- Quantified the cost of dense sums on programmable cores: Flare peaks at 54% of its 6.4 Tbps line rate and needs an estimated $10.6\times$ more logic area than ideal fixed adders at line rate; instruction traces attribute 73% of handler cycles to loads and stores.

Demonstration System for P4Auth, IIT Hyderabad 2025

- Built the live demonstration for P4Auth, showing three configurations side by side: baseline HULA load balancing, HULA under a compromised switch, and HULA protected by P4Auth.
- Acknowledged in *Securing In-Network Traffic Control Systems with P4Auth* (IEEE/IFIP DSN 2025). [IEEE Xplore]

Programmable Data Plane Security (P4TVal), IIT Hyderabad Aug 2024 – Apr 2025

- Implemented Rule Enforcement Verification (REV), the state-of-the-art baseline, in a BMv2–Mininet testbed using P4; simulated adversarial table-rule modification via the switch CLI.
- Measured attack-detection time against network size for both systems, establishing that P4TVal detects rule manipulation $10\times$ faster than REV and stays constant as hop count grows.
- Evaluated data-plane overhead on an Aurora 610 Intel Tofino switch (bf-sde 9.9.0, PTF): ingress-pipeline latency distribution over 5,000 packets, showing $\sim$50 ns added per packet. [code]

Cybersecurity Research Intern, NIT Surathkal Jun 2024 – Jul 2024

- Designed a flexible XOR-based secret image sharing scheme enabling partial recovery with perfect secrecy; published as XF-SISS at IEEE OCIT 2025.
- Evaluated PSNR benchmarks over large image datasets using Python, OpenCV, and NumPy.

Projects

BoSy — Reactive Synthesis Tool Extensions, IIT Hyderabad — course project, team of 3 Jan – May 2026

- Added AIGER circuit output and integrated the Pedant DQBF solver into BoSy, an open-source bounded-synthesis tool written in Swift.
- Authored the technical documentation for the extended toolchain, covering solver backends, bound-search strategies, and a guide to the codebase. [code]

Teaching

Teaching Assistant, Network Security (CS6903) — IIT Hyderabad Jan – May 2026

- Graduate-level course, with Dr. Praveen Tammana and Dr. Saurabh Kumar.
- Designed capture-the-flag problems, quizzes, and assignments; conducted vivas, graded submissions, and invigilated examinations.

Teaching Assistant, Computer Networks (CS3530) — IIT Hyderabad Jul – Nov 2025

- Undergraduate course, with Dr. Praveen Tammana.
- Prepared quizzes and assignments, graded submissions, and invigilated examinations.

Industry Experience

GenAI Intern, Mondee Tech May 2025 – Jul 2025

- Built a multi-agent system with live voice interaction over WebSockets using Gemini Live; integrated LLMs, VLMs, STT/TTS, and OCR models into agent workflows.
- Engineered FastAPI backends and deployed them to production.

Technical Skills

Languages: C++, C, Python, P4, Bash, SQL

Networking & Systems: Intel Tofino, BMv2, Mininet, P4Runtime, PTF, DPDK, RDMA, Linux, gRPC, Git

ML / AI: vLLM, LLMs, VLMs, OCR, agent workflows, LangGraph, Hugging Face

Other: FastAPI, WebSockets, PostgreSQL, MySQL, OpenCV, NumPy, Google Cloud